Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. The learning rate was set to 0.0003 with the Adam optimizer. We trained each model for 30 epochs with a batch size of 32.

2 Results

Figure 1 shows the validation accuracy during training.

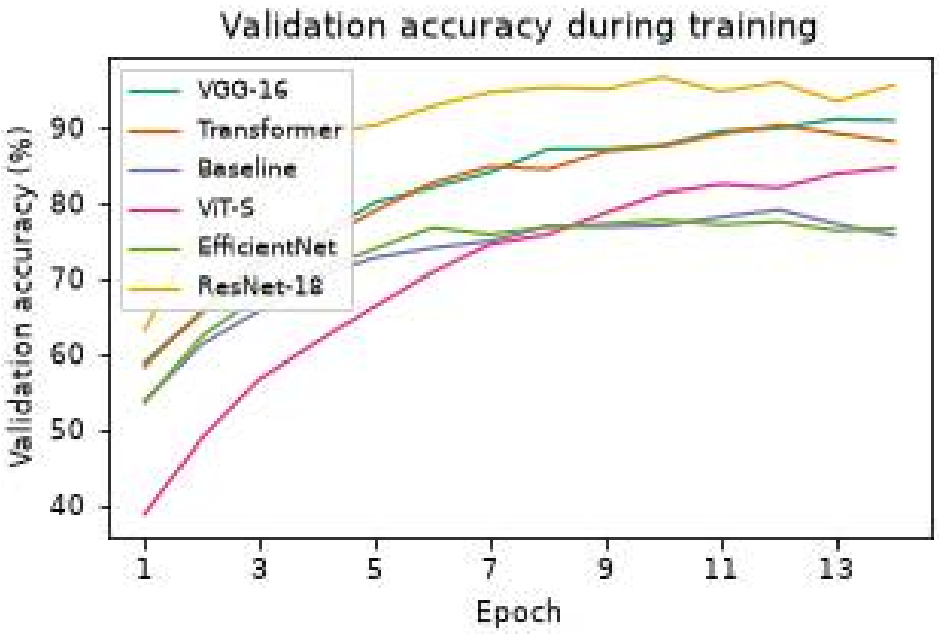


Figure 1: Validation accuracy during training for the models in this experiment.

At epoch 14, VGG-16 achieves 87.7% validation accuracy. Baseline attains a validation accuracy of 76.6% at epoch 4.

Figure 2 shows the accuracy of GRU by dataset.

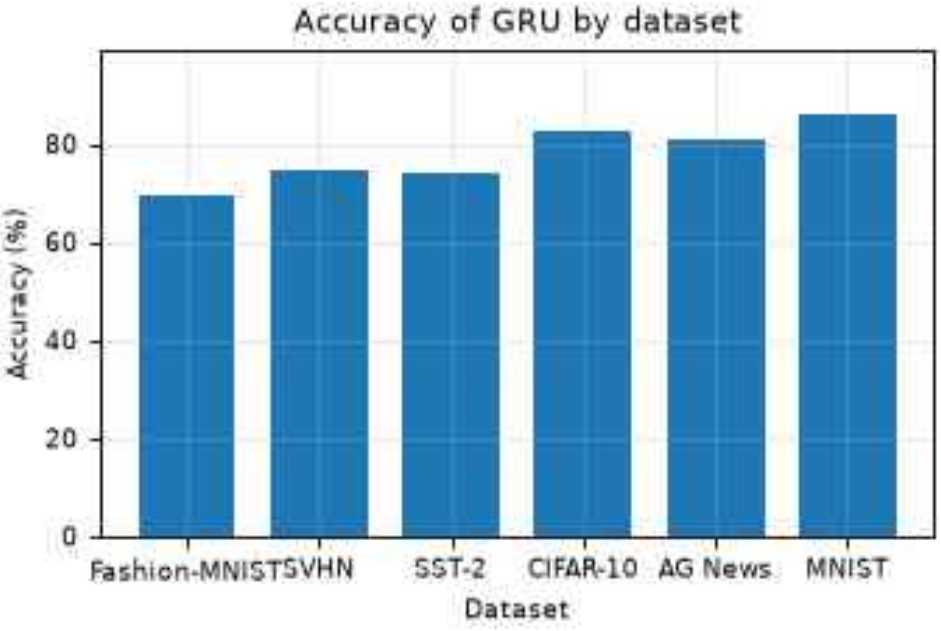


Figure 2: Accuracy of GRU by dataset for the models in this experiment.

As shown in Figure 2, GRU achieves 76.1% accuracy on AG News. As shown in Figure 2, GRU achieves

over 70% accuracy on SST-2. GRU achieves 94.4% accuracy on SVHN.

Figure 3 shows the validation accuracy by dataset.

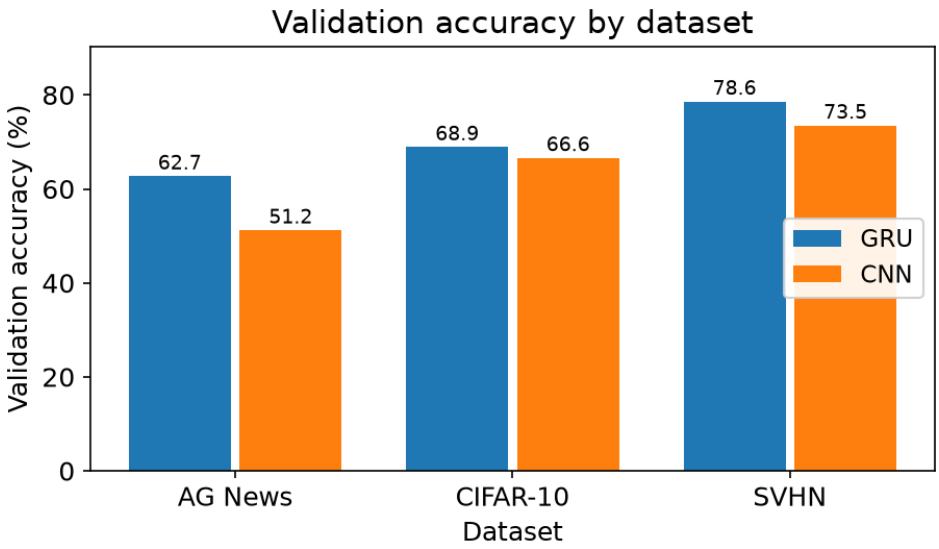


Figure 3: Validation accuracy by dataset for the models in this experiment.

On SVHN, CNN obtains a validation accuracy of about 70%. CNN achieves 6.0% validation accuracy on AG News. As shown in Figure 3, GRU achieves 58.2% validation accuracy on AG News.

3 Discussion

On QQP, Logistic Regression obtains an accuracy of 66.1%. Overall, the results are in line with what we expected from the literature.